

Ruslan V. Smetanin

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Education

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| 2023 – present
Moscow, Russia | PhD
<i>Lomonosov Moscow State University (MSU)</i> 
PhD in Bioinformatics. Faculty of Biology, Department of Microbiology.
Bioinformatic analysis of microbial communities, sequencing data analysis, working with bioinformatic pipelines.
Advisor: Prof. Bonch-Osmolovskaya Elizaveta Aleksandrovna, Corresponding Member of the Russian Academy of Sciences. |
| 2021 – 2023
St. Petersburg, Russia | Master's degree
<i>Alferov University (SPBAU RAS)</i> . 
Master's degree in Physics. Department of Nanobiotechnology.
MSc thesis: Identification of new BRICHOS antimicrobial peptides in genomic data and their structural and functional studies.
Advisor: Prof. Shamova Olga Valeryevna, Corresponding Member of the Russian Academy of Sciences.
Dr. Eliseev Igor Evgenievich. |
| 2016 – 2021
St. Petersburg, Russia | Bachelor's degree
<i>St. Petersburg State Agrarian University (SPBGU)</i> . 
Bachelor's degree in Agrosoil Science. Department of Agricultural soil science and ecology.
BSc Thesis: The influence of bacterial fertilizers on the biological activity of sod-podzolic soil and the yield of oats in the conditions of the Leningrad region.
Advisor: Associate Prof. Gamzaeva Rufina Sidmetovna. |

Professional Experience

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| 2024 – present
Moscow, Russia | Bioinformatics Analyst
<i>Lomonosov Moscow State University (MSU)</i>
Bioinformatic analysis of microbial communities, sequencing data analysis, working with bioinformatic pipelines |
| 2023 – 2023
Moscow, Russia | Internship (laboratory researcher)
<i>Shemyakin–Ovchinnikov Institute of bioorganic chemistry RAS.</i>
Responsibilities: Molecular dynamic of protein, bioinformatic analysis structure of proteins using GROMACS and PyMol. Protein extraction, protein purification (chromatography column, HPLC), western blotting.
Advisor: Dr. Bychkov Maxim Leonidovich. |
| 2022 – 2023
St. Petersburg, Russia | Technician (laboratory researcher)
<i>Renewable energy laboratory of Alferov University</i> 
Responsibilities: Bioinformatic analysis of RNAseq using Python/R/Linux and visualization of proteins in PyMol and molecular docking in AlfaFold2.
Advisor: Dr. Eliseev Igor Evgenievich. |
| 2021 – 2021
St. Petersburg, Russia | Internship (laboratory researcher)
<i>Institute of Experimental Medicine of the National Medical Research Center. V. A. Almazova</i>
two-month internship.
Responsibilities: Analysis of Sanger sequencing data. |
| 2017 – 2019
St. Petersburg, Russia | Technician (laboratory researcher)
<i>All-Russian Scientific Research Institute of Agricultural Microbiology</i> 
Responsibilities: Growing alfalfa plants (~500 plants), preparing soil for plants, crushing soil, statistical analysis of data and writing the article.
Advisor: Dr. Yakobi Lidia Mikhailovna. |

Publications

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| 2024 | First vertebrate BRICHOS antimicrobial peptides: β-hairpin host defense peptides in limbless amphibia lung resemble those of marine worms. (Q1) 
<u>SMETANIN R.V.</u> ; SUKHAREVA M.S.; VLADIMIROVA E.V.; ZHARKOVA M.S.; MIKUSHINA A.D.; KOMLEV A.S.; KHAYDUKOVA M.M.; FILATENKOVA T.A.; KALGANOVA A.I.; PIPIYA S.O.; TEREKHOV S.S.; ORLOV D.S.; SHAMOVA O.V.; ELISEEV I.E.
DOI: 10.1016/j.bbrc.2024.149913 |
| 2020 | Study of the structural and functional organization of arbuscular mycorrhiza in the original and mutant lines of hop alfalfa (medicago lupulina) with different methods of inoculation with the fungus rhizophagus irregularis. Mikologiya i Fitopatologiya. 
ZHELEZNYAKOV S.V.; LEBEDEV V.K.; <u>SMETANIN R.V.</u> ; JAKOBI L.M.
DOI: 10.31857/S0026364820030125; (in Russian). |

Presentations

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| 2024
St. Petersburg, Russia | Unexpected discovery of β-hairpin amp in legless amphibians: a new hope in the fight against multidrugresistant pathogens
<i>Institute of Experimental Medicine</i>
Book of Abstracts IV of Mini-Conference with International Participation. 2024 |
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SMETANIN R.V., SUKHAREVA M.S., VLADIMIROVA E.V., ZHARKOVA M.S., MIKUSHINA A.D., KOMLEV A.S., KHAYDUKOVA M.M., FILATENKOVA T.A., KALGANOVA A.I., PIPIYA S.O., TEREKHOV S.S., ORLOV D.S., SHAMOVA O.V., ELISEEV I.E.
DOI: 10.18720/SPBPU/2/id24-381

- 2024
St. Petersburg, Russia
IV International Mini-Conference "Antimicrobial Peptides as prototypes of novel antibiotics", (2024). 
Oral presentation: Unexpected discovery of β -hairpin AMP in legless amphibians: a new hope in the fight against multidrug-resistant pathogens
SMETANIN R.V.; SUKHAREVA M.S.; VLADIMIROVA E.V.; ZHARKOVA M.S.; MIKUSHINA A.D.; KOMLEV A.S.; KHAYDUKOVA M.M.; FILATENKOVA T.A.; KALGANOVA A.I.; PIPIYA S.O.; TEREKHOV S.S.; ORLOV D.S.; SHAMOVA O.V.; ELISEEV I.E.
- 2022
St. Petersburg, Russia
XXII Winter Youth School on Biophysics and Molecular Biology. (2023), 190-191 
St. Petersburg Institute of Nuclear Physics named after B.P. Konstantinov National Research Center "Kurchatov Institute" together with the Kurchatov Genomic Center
Poster Presentation: Identification and Structural-Functional Studies of New BRICHOS Antimicrobial Peptides in Genomic Data, Published in the Collection of Abstracts; **(in Russian).**
- 2021
St. Petersburg, Russia
Dynamics of activity of urease and rhizosphere microflora in otogeny against the background of the use of biological products. Innovative ideas of young researchers for the agro-industrial complex. (2021), 130-133. 
GAMZAEVA R.S.; BASHARINA M.V.; **SMETANIN R.V.**
eLIBRARY ID: 46349690; **(in Russian).**
- 2020
Moscow, Russia
Winter Shcool
Moscow Institute of Physics and Technology (MIPT)
Biological section: DNA extraction, RT-PCR, gel electrophoresis, western blotting.
- 2020
St. Petersburg, Russia
Winter School
Alferov University (SPBAU RAS)
Targeting human oncogenic receptor ErbB3/HER3 with a single-damain antibody. Expression of recombinant proteins in *E. coli*, high-performance liquid chromatography (HPLC), dialysis, HeLa cell cultivation, immunocytochemical staining of cancer cells, confocal microscopy.
- 2019
St. Petersburg, Russia
Study of the induced mutant of hop alfalfa by symbiotic traits during the formation of mycorrhizal symbiosis. International Congress "VII Congress of the Vavilov Society of Geneticists and Breeders". (2019), 930.
St. Petersburg University (SPBU)
Poster Presentation: Study of the induced mutant of hop alfalfa (*medicagolupulinal. subsp. vulgariskoch*) by symbiotic traits during the formation of mycorrhizal symbiosis.
JACOBI L.M.; ZHELEZNYAKOV S.V.; **SMETANIN R.V.**; LEBEDEVA V.K.; KOZHEMYAKOV A.P.
eLIBRARY ID: <https://www.elibrary.ru/item.asp?id=39157063>; **(in Russian).**

Awards

- 2022
Presidential Scholarship for Young Scientists.
Persons who have shown outstanding abilities and have shown high achievements in a particular field of activity.
- 2022
St. Petersburg Government Scholarship.
Competition for Outstanding Students of Higher Education Institutions for the Right to Receive Named Scholarships from the Government of St. Petersburg.
- 2024
Patent
Number of patent: RU 2826459 C1
СИНТЕТИЧЕСКИЕ АНАЛОГИ ПРИРОДНЫХ ПЕПТИДОВ, ОБЛАДАЮЩИХ АНТИБАКТЕРИАЛЬНОЙ АКТИВНОСТЬЮ

Skills

Molecular biology:

DNA and protein extractions, PAGE gels, Reaction of restriction, PCR, western-blotting, high-performance liquid chromatography (HPLC).

Genetic Engineer:

Development and production of genetically engineered constructs, proficiency in fundamental cloning methods, and de novo synthesis of genetic sequences.

Bioinformatics:

Data Analysis | NGS, Snakemake, SnapGene, IGV, PyMOL, Qiime, AlphaFold2.

Databases | Kraken2, Silva, GTDB, PDB.

Programming | Python, R, bash.

Cell biology:

Cell culture cultivation HEK293f, transfection, obtaining membrane protein fraction, flow cytometry.

Volunteering

Memberships & Affiliations

Member of the Microbiological Community

Member of the All-Russian Society of Geneticists and Breeders (VOGiS)

Languages

English

Intermediate, B1

